Logan Luna

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RESEARCH INTERESTS

Deep learning theory and optimization; learning dynamics and gradient-based methods; representation learning and inductive bias in neural architectures; stable and interpretable learning mechanisms; reinforcement learning in safety-critical and real-world systems

EDUCATION

Georgia Institute of Technology — Atlanta, USA
Master of Science: Computer Science — August 2026 — Expected December 2027
Specialization: Artificial Intelligence — Track: Thesis

Embry-Riddle Aeronautical University — Daytona Beach, USA
magna cum laude — Honors Program — Outstanding Undergraduate in Computer Science — August 2022 — May 2026
Bachelor of Science: Computer Science — Minor: Applied Mathematics — Cumulative GPA: 3.85/4.00

RESEARCH & PROFESSIONAL EXPERIENCE

Naval Surface Warfare Center Dahlgren Division (NSWCDD)
Scientist Intern, Department of Defense (Beam and Optic Control for High Energy Weapons) — 2024 — Present
Active Security Clearance Level: Secret

- Designed a lightweight temporal feature approach to improve stability when targets appear infrequently, under deployment constraints that limited heavier temporal models.
- Built a real-time object tracking pipeline using YOLO, ORB keypoint matching, homography-based optical flow, re-identification, and Kalman filtering.
- Integrated ML/CV with additional sensing inputs (e.g., radar signals) to improve detection precision and track continuity in dynamic environments.

Reasoning and Learning Group — Georgia Institute of Technology
Researcher (AI4Math / Auto-formalization) — 2025 — Present
Group Lead: Dr. Vijay Ganesh

- Contribute to *LogicComplexityLib*, a project aimed at auto-formalizing logic and model theory textbooks using neuro-symbolic reasoning tools.
- Work to extract and structure textbook theorem–proof pairs and evaluate Aristotle-generated Lean formalizations.
- Analyze type-checking failures, missing dependencies, and semantic mismatches to support dataset quality and tooling refinement.

Agile Research Group — Embry-Riddle Aeronautical University
Researcher (Satellite Trajectory Optimization) — 2024 — Present
Group Lead: Dr. Omar Ochoa

- Developed an RL-based framework (PPO) for fuel-aware satellite collision avoidance in custom astrodynamics simulation.
- Studied reward shaping and curriculum learning to balance collision avoidance, fuel conservation, and orbital stability.
- Contributed to reproducible evaluation via deterministic replay and telemetry logging.

PUBLICATIONS

Journal Articles

- **Logan Luna**, Juan Ortiz Couder, Raul Alejandro Vargas-Acosta. *Satellite Trajectory Optimization via Proximal Policy Optimization for Space Debris Avoidance. IEEE Access*, 2026. DOI: 10.1109/ACCESS.2026.3655237

Peer-Reviewed Conference Papers

- **Logan Luna**, Matthew P. Berkowitz, Laxima Niure Kandel, Sirio Jansen-Sánchez. *Dueling Deep Q-Learning for Intrusion Detection.* IEEE SoutheastCon, 2025. **Best Paper Award (IEEE–Eta Kappa Nu)**. DOI: 10.1109/SoutheastCon56624.2025.10971436
- **Logan Luna**, Sirio Jansen-Sánchez, Ilteris Demirkiran, Leo Ghelarducci. *Signal-Centric Remote Sensing via Alternative Preprocessing and Acoustic Processing for ML-Driven Applications.* IEEE SoutheastCon, 2025, Concord, NC, USA. DOI: 10.1109/SoutheastCon56624.2025.10971547
- **Logan Luna**, Joseph Rigo, Kellan Shew, Raul Alejandro Vargas-Acosta. *SensorWF: A FAIR Generalizable Workflow Framework for Scientific Time-Series Analysis.* Under Review, IEEE International eScience Conference, 2026.

LEADERSHIP & SERVICE

Autonomous Maritime Robotics Association (AMRA) — RoboSub Team
Software Department Head 2022 — Present

- Lead AUV autonomy software integrating computer vision, sonar processing, and ML-based decision-making.
- Built training/onboarding processes and a supportive team culture to improve retention and development speed.
- Support mentoring and research development within the team; contributed to a 16th-place finish out of 52 schools at international competition.

GRANTS & FUNDING

Total funding secured: $27,960

- **Center for Aerospace Resilient Systems (CARS)**. Research Grant ($24,000), Spring 2026.
  AI for Data Anomaly Detection in a Satellite Learning Laboratory (SATLL) (Project Lead).
- **Embry-Riddle Aeronautical University — Office of Undergraduate Research**. Spark Grant (up to $600), 2025.
  Conference travel support for IEEE SoutheastCon 2025 (Project Lead).
- **Embry-Riddle Aeronautical University — SIG Research Project Grant**. Research Grant ($3,360), 2024–2025.
  Signal-Centric Remote Sensing via Alternative Preprocessing and Acoustic Processing for ML-Driven Applications (Project Lead).

SELECTED PROJECTS

AI for Data Anomaly Detection in a Satellite Learning Laboratory
Project Lead 2025 — 2026
Externally funded research project

- Lead development of ML-based anomaly detection pipelines for satellite telemetry streams in the Satellite Learning Laboratory (SATLL).
- Design data-centric preprocessing workflows operating directly on structured telemetry (CSV/HDF5), avoiding image-based representations to reduce onboard computation.
- Implement and evaluate baseline (z-score) and advanced models (Isolation Forests, autoencoders) for multivariate anomaly detection across power, thermal, attitude, and communications subsystems.

Reinforcement Learning for Aerial Robotics: Sim-to-Real Transfer
Project Member (Flexible & Intelligent Complex Systems Research Group) 2025 — 2026

- Train RL control policies for Crazyflie 2.x nano-quadrotors in NVIDIA Omniverse Isaac Gym.
- Evaluate sim-to-real gaps (stability, drift, tracking) under perturbations (noise, wind, mass variation).

Enhancing Underwater Object Recognition with Computer Vision and Filtering Techniques
Project Lead 2023 — 2026

- Use YOLOv8 for detection/classification and feature matching pipelines (SuperPoint/SuperGlue) for AUV perception.
- Combine Kalman predictions with detections for continuous tracking; plan confidence-aware fusion across model outputs.

AWARDS

- Embry-Riddle's Outstanding Undergraduate in Computer Science (2025 & 2026)
- IEEE–Eta Kappa Nu Best Paper Award (2025)
- DoW SMART Scholarship (2023 — Present)
- 1st Place, Embry-Riddle Student Research Symposium (2023)

SKILLS

Programming

- Python, C/C++, Java

ML/CV

- PyTorch, TensorFlow, OpenAI Gym, SHAP, OpenCV, YOLO

Mathematics

- Linear Algebra, Calculus I–III, Differential Equations, Probability & Statistics, Discrete Mathematics, Optimization

Tools

- Git, Docker, Linux, NVIDIA Isaac Lab, Lean

REFERENCES

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